

Nandakumar Vuppalapati

Data Engineer - Python, AWS, Databricks, dbt, Spark

Tampa, FL • 424-366-6034 • nandakumarvuppalapati@gmail.com • linkedin.com/in/v-nandakumar

Professional Summary

Data Engineer with progressive experience building and supporting production ETL/ELT pipelines, distributed data processing, and cloud warehouse/lakehouse platforms across financial services and healthcare. Hands-on across Python, SQL, Spark/PySpark, Databricks, dbt, Airflow, Snowflake, AWS, and Azure, with experience spanning ingestion, dimensional modeling, testing, observability, governance, and production support. Recent work extends into ML feature pipelines, MLflow-supported workflows, and LLM/RAG tooling for regulated data and document use cases. Comfortable owning data workflows end to end - from ingestion and transformation through validation, monitoring, troubleshooting, and downstream delivery.

Technical Skills

Languages: Python (Pandas, PySpark), SQL (window functions, CTEs, query tuning), T-SQL, Bash

Cloud: AWS (S3, Glue, Lambda, Step Functions, Redshift, Athena), Azure (Databricks, Data Factory, Synapse Analytics, Blob Storage/ADLS Gen2, Azure SQL)

Data Processing: Apache Spark, PySpark, Databricks, Spark Structured Streaming, Apache Kafka

Orchestration & Transformation: Apache Airflow (DAGs, scheduling, sensors, retries, failure handling), dbt, Azure Data Factory

Warehouse / Lakehouse: Snowflake, Databricks, Delta Lake, AWS Redshift, Azure Synapse; Medallion Architecture (Bronze/Silver/Gold)

Databases: Oracle, PostgreSQL, MySQL, Azure SQL, Databricks SQL, Pinecone (vector database)

DevOps / IaC: Docker, Terraform, GitHub Actions (CI/CD), Git

Data Reliability & Governance: Great Expectations, dbt tests, data contracts, data lineage, PII masking, RBAC, SOX-aware controls, compliance documentation

AI / ML Data Systems: MLflow, feature engineering, XGBoost, RAG, vector retrieval, LangChain, OpenAI API, Pinecone, Feast, Ragas, FastAPI

Work Experience

AI Data Engineer, Workiva, Remote, USA

Dec 2025 - Present

- Built and maintained Python/Airflow financial data pipelines across AWS S3, Glue, Lambda, Step Functions, and Redshift, processing 1M+ daily records across payments, risk, settlement, compliance, and downstream analytics while handling production failures, reruns, and recovery validation.
- Developed Snowflake/dbt transformation layers using reusable SQL models, macros, automated tests, and medallion-style data organization, maintaining 30+ financial models that published governed datasets for reporting, risk analysis, and analytical consumers.
- Built **Spark/Databricks** feature and embedding pipelines over curated financial data for downstream fraud and ML workflows, supporting feature preparation and model lifecycle processes with **MLflow**.
- Implemented **Prometheus/Grafana** monitoring across critical **ETL** workflows, tracing failures through orchestration, transformation, and downstream loading stages and validating data consistency before pipelines returned to normal schedules.
- Built a separate regulatory-document RAG proof of concept with LangChain and OpenAI APIs, combining chunking, embeddings, and retrieval so compliance users could search regulatory content in natural language.

Data Engineer, Wellmark BCBS, Des Moines, USA

Aug 2024 - May 2025

- Built Azure Data Factory ingestion pipelines to consolidate 10M+ records from 30+ fragmented legacy healthcare sources through Blob Storage and Azure SQL, standardizing source-to-target processing and applying PII handling controls before downstream transformation.
- Developed **PySpark** transformation jobs in **Azure Databricks** to clean, standardize, join, and curate data across medallion-style layers for reporting and analytics, replacing portions of legacy SQL-heavy processing with distributed transformations.
- Created 15+ SQL/dbt transformation models with dbt schema tests and Great Expectations checkpoints, catching schema, null, duplicate, and integrity issues before curated data reached downstream consumers.
- Maintained metadata and lineage documentation and supported repeatable platform delivery with Terraform, Docker, and GitHub Actions, keeping data changes traceable across development and deployment.
- Built an LLM-assisted search prototype across a 1,000+ document vendor-contract repository using document extraction, embeddings, vector retrieval, and natural-language Q&A; evaluated MCP as a possible pattern for broader enterprise document access.

Junior Data Engineer, Sigmoid, Bengaluru, India

Jun 2022 - Dec 2023

- Refactored **T-SQL** stored procedures and indexing strategies behind recurring finance-reporting workloads, resolving long-running queries and recurring month-end reporting bottlenecks.
- Built a **Python**-based validation framework with **pytest** checks integrated into **ETL** processing, catching schema and null-value issues before data reached stakeholder reports.
- Redesigned a flat reporting structure into a **star-schema** dimensional model in **Azure Synapse**, organizing analytical data around fact and dimension tables for more maintainable finance reporting.
- Automated a manual Excel-based weekly reporting workflow with scheduled Azure Data Factory pipelines sourcing directly from Azure SQL, removing repeated spreadsheet preparation and saving roughly 6 hours of manual work each week.
- Supported scheduled ADF pipelines in production, investigated failed runs, validated outputs after recovery, and managed Blob Storage lifecycle policies for staging data.
- Configured ADF schedule triggers and failure alerts and handled recurring data requests from finance analysts, including reruns and output checks when upstream data arrived late.

Data Scientist, Tech Citi Software, Bengaluru, India

Jul 2021 - Dec 2021

- Migrated historical loan and transaction data from **Oracle** to **PostgreSQL**, resolving schema and datatype mismatches and validating migrated records for consistent downstream reporting and analysis.
- Built Python and Apache Airflow ETL workflows to ingest 50,000+ daily financial transaction records from CSV and REST API sources into PostgreSQL on a repeatable schedule, replacing manual data-loading steps for downstream reporting and modeling.
- Engineered transaction-level behavioral features and trained an offline XGBoost fraud-classification prototype with SMOTE to address class imbalance, reaching an AUC-ROC of 0.85 in historical-data evaluation.

Projects

Real-Time ML Feature Platform for Fraud Scoring

Kafka, Flink, Feast, Databricks, MLflow, FastAPI, Docker, AWS

- Built to close the gap between batch fraud pipelines and low-latency scoring: Kafka handles event ingestion, Flink computes streaming features, Feast keeps online and offline features aligned, and MLflow/FastAPI support model versioning and serving. The design focuses on feature freshness and training-serving consistency.
- Kept feature computation, feature storage, and serving as separate concerns so the pipeline can evolve without tightly coupling the streaming and API layers; containerized the services for repeatable testing.

CDC Lakehouse with Data Contracts

Debezium, Kafka, Apache Iceberg, dbt Core, Great Expectations, PostgreSQL, Docker

- Built to catch breaking changes close to the source: Debezium captures PostgreSQL row-level changes, Kafka transports them into Iceberg, and dbt plus Great Expectations handle transformation and validation. A schema-contract layer fails incompatible changes before downstream consumers are affected.
- Designed contract checks around schema compatibility and required fields so breaking source changes fail visibly instead of silently propagating into downstream models; containerized the stack to reproduce CDC and recovery scenarios.

RAG Evaluation & Observability Harness

LangChain, OpenAI API, Pinecone, Ragas, FastAPI, Docker, Airflow

- Built around a practical production question - how do you know when a RAG system is getting worse? A golden-question dataset evaluates retrieval and generation separately, with Ragas checks and scheduled Airflow runs surfacing retrieval failures and hallucination regressions.
- Tracked retrieval and generation separately so a weak answer can be traced to missing context versus generation behavior, making the harness useful for regression testing instead of relying on a single aggregate score.

Education

Master of Computer and Information Sciences - University of South Dakota, Vermillion, SD, USA

2025

Bachelor of Engineering - Indian Institute of Information Technology (IIIT), India

2023

Certifications

AWS Certified Cloud Practitioner - Amazon Web Services

2025